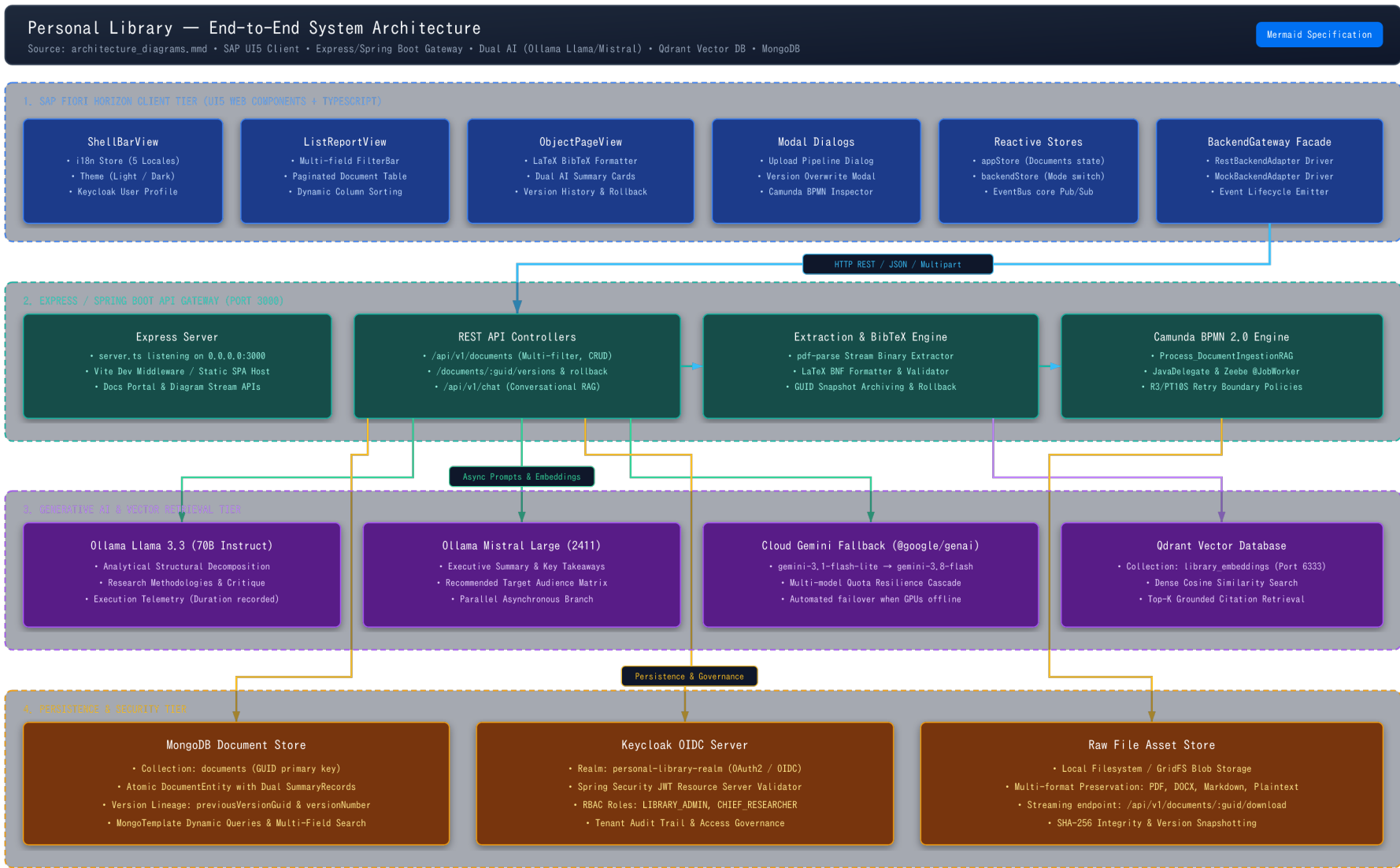


Personal Library · End-to-End System Architecture Specification

Specification Model: architecture_diagrams.mmd · SAP Fiori UI5 · Express Gateway · Dual AI · Qdrant · MongoDB



Architectural Tiers & Component Catalog

Detailed breakdown of layers mapped from the Mermaid graph TD definition

1. Client Tier (UI5)

- ShellBarView
Navigation, branding, theme switch, i18n store.
- ListReportView
Multi-field FilterBar, sorting, table pagination.
- ObjectPageView
BibTeX editor, dual AI cards, citation RAG chat, version history & rollback matrix.
- Modal Dialogs
Upload pipeline, version overwrite, rollback modal, BPMN inspector.
- Observable Stores
appStore, backendStore, themeStore, i18nStore.
- BackendGateway
Pluggable driver facade for REST & Mock modes.

2. API Gateway (Port 3000)

- Express Server
Port 3000 on 0.0.0.0, dev Vite middleware, static host.
- REST Controllers
/documents, /versions, /rollback, /chat, /auth.
- Extraction Engine
pdf-parse binary stream extractor, LaTeX BNF formatter.
- Version Control & Rollback
In-place version overwrite, snapshot archiving, non-destructive rollbacks.
- Camunda BPMN Engine
Execution delegates & Zeebe workers for RAG pipeline.

3. AI & Vector Retrieval

- Ollama Llama 3.3 (70B)
Analytical decomposition, methodology critique, data highlights.
- Ollama Mistral Large
Executive summary, conclusions, audience recommendations.
- Gemini Cloud Fallback
@google/genai cascade fallback when GPU offline.
- Qdrant Vector DB
Dense cosine similarity retrieval (library_embeddings).
- Semantic Chunking
~350 char boundaries with paragraph context.

4. Persistence & Security

- MongoDB Store
Collection "documents", GUID primary key, version lineage.
- Keycloak OIDC Server
Realm: personal-library-realm, OAuth2 JWT bearer.
- RBAC Governance
LIBRARY_ADMIN and CHIEF_RESEARCHER roles.
- Raw Asset Store
PDF, DOCX, Markdown, streaming downloads.
- Audit Logging
Complete tenant event tracing & security trail.